# Tibial Plateau Fracture Protocol Phase 1

## Purpose

**Maximum Protection**

## Typical Time Frame

**0-6 Weeks**

## Precautions

- NWB w/ crutches for 6 weeks
- If patient had meniscus repair limit ROM to 60°
- Protect fixation
- Increase knee flexion ROM gradually by 10°-15° each week

## Criteria to Pass

- 0°-90° of knee AROM
- 30 consecutive SLR without lag

## Priority List & Suggested Exercises

1. **Knee Flexion ROM**
   a. Seated knee bend off table, Physioball knee flexion (supine), Heel slides (seated and supine), Wall slides
   b. Patella mobs (inferior glides)
2. **Quad Activation (at least 10min of NMES between 1-2 exercises)**
   a. Quad sets with NMES (towel roll, 10sec hold)
   b. SLR with NMES (towel roll > quad set > into SLR > reset)
   c. Kicking into physio ball isometric holds (begin at 90° and progress to 60°) NMES if necessary
3. **Knee Extension ROM**
   a. Low Load Long Duration: Extension prop, Foot on Ottoman (add weight or ice to all)
   b. Manual Techniques: Passive extension overpressure (add quad set), Patella mobs (superior glides)
4. **Lateral and Posterior Chain**
   a. 4-way SLR with (NMES only for forward)
   b. Clamshells
   c. Standing hamstring curl (hamstring curl with surgical side)
   d. Straight leg bridge on ball

## Special Considerations

- Manage pain with meds
- Swelling: compression, elevation, ice, retrograde massage
- NMES: Use with all quad exercises in this phase
- BFR: use as soon as patient able to get a good quad contraction
- Incision: inspect frequently to ensure healing and closing
- Consider aerobic conditioning for HEP throughout all phases to maintain aerobic capacity

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# Tibial Plateau Fracture Protocol Phase 2

## Purpose

**Protection**

## Typical Time Frame

**6-12 Weeks**

## Precautions

- WBAT to full WB with no AD by 10-12 weeks
- Increase amount of weightbearing by ~ 25% each week until full weight bearing

## Criteria to Pass

- Knee ROM is restored to 80% of the uninvolved knee
- SLS x 30sec without LOB
- Normal gait pattern without AD and with full terminal knee extension

## Priority List & Suggested Exercises

1. **Knee Flexion ROM**
    a. Continue Phase 1 exercises (adding sets and reps per discretion)
    b. Nu-step & bike (½ revolutions progressing to full)
2. **Quad Activation**
    a. 4-way SLR (NMES only for forward)
    b. TKEs with NMES progression
    c. Leg press into physioball with NMES
    d. SAQ with NMES progression
    e. LAQ with NMES
3. **Gait Training**
    a. SLS no UE support > SL balance reaches
    b. Gait training without crutches (forward and retro) > progress to sled (pulling and pushing)
    c. Low hurdle (reciprocal pattern) > high hurdle (step to) > high hurdle (reciprocal pattern)
    d. Calf Raises
4. **Knee Extension ROM**
    a. Continue Phase 1 exercises (adding sets and reps per discretion)

## Special Considerations

- Swelling: compression, elevation, ice, retrograde massage
- NMES: Use for at least 10 minutes until patient can perform unweighted SAQ and LAQ comparable ROM to uninvolved LE
- BFR: use as soon as patient able to get a good quad contraction (start at 50% and progress to 80% per patient tolerance and use most of session)
- Incision: Should be closed by 6 weeks
- Programming: High volume for exercises 2-3 sets of 12-15 reps

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# Tibial Plateau Fracture Protocol Phase 3

## Purpose

**Functional Restoration**

## Typical Time Frame

**8-14 Weeks**

## Precautions

- Avoid exercises that are painful and cause swelling
- Don’t initiate running
- Still avoid aggressive PROM into knee flexion

## Criteria to Pass

- Knee ROM is restored to 90% of the uninvolved knee in supine
- Ascend/descend stairs reciprocal pattern without pain or UE assist
- 5 steps ups (10" step)
- 5 lateral step downs (8" step)
- Squat symmetrically to 90°
- Y balance reach 80% of uninvolved side without hip strategy

## Priority List & Suggested Exercises

1. **Knee Flexion ROM**
   a. Continue Phase 2 exercises (adding sets and reps per discretion)
   b. Manual: Knee flexion with manual tibial internal rotation
   c. Mobility: Prone quad stretch, Knee Circles
   d. AAROM: Shuttle press, TRX squat
2. **Anterior Chain Strength**
   a. Continue SAQ and/or LAQ with NMES and isometric knee extension at 60°
   b. Isotonic leg extension (ankle weight > cable resistance)
   c. Wall sit progression
   d. Heavy banded TKEs/Shuttle Press TKEs
   e. SL Shuttle press
3. **Functional Strength**
   a. Squat progression
   b. Step ups progression
   c. LSD progression
4. **Posterior Chain Strength**
   a. Bridge progression
   b. Hamstring curls progression
   c. Quadruped hip extension over bench holds
   d. Double leg hip thruster holds
   e. SL RDL reaches
   f. Calf raise variations
5. **Lateral Chain Strength**
   a. Clamshells > Side lying abduction
   b. Stork stance holds
   c. Sidelying shuttle press
   d. Hip hike holds > hip hike reps

## Special Considerations

- BFR: Continue to use in the earlier part of PT session
- NMES: Use for at least 10 minutes until patient can perform unweighted SAQ and LAQ = ROM to uninvolved LE
- Motor Control: If there is still a difference between AROM/PROM focus on end range strengthening and isometrics
- Programming:
  - High Volume Exercises 2-3 sets of 12-15 reps
  - 3:1:1 anterior chain to posterior chain to lateral chain strengthening ratio
  - Complete 1x25 reps on uninvolved leg for anterior chain exercises and same sets/reps for posterior and lateral chain

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# Tibial Plateau Fracture Protocol Phase 4

## Purpose

**Progressive Overload**

## Typical Time Frame

**12-18 Weeks**

## Precautions

- Avoid exercises that are painful and cause swelling
- Don’t initiate running

## Criteria to Pass

- Knee ROM is restored to 95% of the uninvolved knee
- Y balance reach 95% of uninvolved side without hip strategy in supine
- 30 lateral step downs (8" step, back flat against wall) with no hip compensation
- 15 forward step downs (8" step)
- 25 SL bridges on bench (knee bent at 20°)
- 25 SL calf raises (10° incline to 60bpm metronome)
- 25 SL squats to box (20") with minimal trunk flexion

## Priority List & Suggested Exercises

1. **Knee Flexion ROM**
   a. Continue previous PROM & AAROM exercises
   b. Tibial rotations
   c. Couch Stretch, Prone quad stretch
2. **Anterior Chain Strength**
   a. DL machine extension holds at 0° > SL holds
   b. DL extension > SL extension
   c. Sled Pulls/Pushes
   d. Kickstand Squat at wall > Goblet squat
   e. Eccentric Shuttle press (up with 2, down with 1)
3. **Functional Strength**
   a. LSD progression
   b. Forward step down progression
   c. SL squat progression
4. **Posterior Chain Strength**
   a. Hip thruster progression
   b. RDL progression
   c. Hamstring curl progression
   d. Calf raise variations
5. **Lateral Chain Strength**
   a. Lateral squat progression
   b. Stork stance banded reps
   c. Lateral band walks/skaters
   d. Side plank progression

## Special Considerations

- BFR: Continue to use in the earlier part of PT session
- NMES: Use for at least 10 minutes until patient can perform unweighted SAQ and LAQ = ROM to uninvolved LE
- Motor Control: If there is still a difference between AROM/PROM focus on end range strengthening and isometrics
- Programming:
  - 3 sets of 6-12 reps with anterior and functional exercises focusing on strength
  - 3:1:1 anterior chain to posterior chain to lateral chain strengthening ratio
  - Tempo: 4-1-2 w/ emphasis on eccentric control
  - Complete 1x25 reps on uninvolved leg for anterior chain exercises and same sets/reps for posterior and lateral chain

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# Tibial Plateau Fracture Protocol Phase 5

## Purpose

**Intro to Impact**

## Typical Time Frame

**14-20 Weeks**

## Precautions

- Avoid exercises that are painful and cause swelling
- Don’t initiate running until patient has passed the criteria below for return to jogging

## Criteria to Pass

**To Begin Jogging: (Use Strengthlevel.com)**

- 12 reps at 70% of Novice Goblet Squats
- 12 reps at 70% of Novice Forward Lunges
- 12 reps at 70% of Novice DB RDLs
- <30% quad strength deficit for average force with Isometric testing at 90° knee flexion
- 15 SL hops in place w/ good load acceptance & landing in same spot

**To Move Onto Phase 6:**

- ATHLETES: Run 1mi w/o limping
- GENERAL POP: Jog 5min w/o limping
- Sprint w/o compensation
- Land symmetrical <10% difference w/ 5 squat jumps on Force Plates
- Decel efficiently in the sagittal plane

## Priority List & Suggested Exercises

1. **Anterior Chain Mobility**
   a. Dip Bar Assisted Slant Board Squat > Slant Board Squat > Dip Bar Assisted exaggerated Split Squat on step > No step > No Dip Bar Assist
2. **Anterior Chain Strength**
   a. Split Squat Isometric Hold (Yoga Block Cue)
   b. DL machine extension holds at 0 Degrees > SL holds
   c. DL machine extension > SL machine extension
   d. Shuttle Press Variations
3. **Functional Strength**
   a. Loaded goblet squat progression
   b. Lunge progression
   c. Loaded step up variations
   d. Loaded LSD & Forward step downs
3. **Agility/Impact**
   a. Agility ladder (basic coordination)
   b. Mini hops > Jogging progression > hop & stop > UE assisted treadmill intervals
   c. Deceleration: snap down progression > cable decels > multiplanar hops > ladder decels > change of pace > stop & goes
   d. Jump progression
4. **Posterior Chain Strength**
   a. Hamstring curl progression
   b. Hip thruster progression
   c. Deadlift/RDL variations
   d. Nordic progression
   e. Calf raise variations
5. **Lateral Chain Strength**
   a. Lateral squat variations
   b. Copenhagens
   c. Anti-rotation and balance training

## Special Considerations

- Stretching: Continue quad stretching (prone quad and couch stretch)
- Quad Strength: Constant focus to improve quad strength and decrease quad deficit
- Isometric Testing: Quad strength testing at 90° to limit anterior knee pain
- Programming:
  - 2:1:1 anterior chain to posterior chain to impact exercises
  - Increase sets to 4 with reps 6-12 specifically for anterior and functional exercises
  - No > 2-3 sets of 3-5 reps for agility/impact exercises
  - Once patient has started to jog begin same sets/reps for all exercises on univolved LE

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# Tibial Plateau Fracture Protocol Phase 6

## Purpose

**Deceleration**

## Typical Time Frame

**5-7 Months**

## Precautions

- Don’t initiate cutting until patient has passed the criteria below for return to cutting

## Criteria to Pass

**To Begin Cutting: (Use Strengthlevel.com)**

- 8 reps at 80% of Intermediate Goblet Squats
- 8 reps at 80% of Intermediate Forward Lunges
- 8 reps at 80% of Intermediate DB RDLs
- <20% quad strength deficit for average force with isometric testing at 90° knee flexion

**To Move Onto Phase 7:**

- SL hop test >80% LSI with average of 3 trials
- <20% deficit for Deceleration RFD and Peak Force with landing w/ SL Countermovement Jump on Force Plates

## Priority List & Suggested Exercises

1. **Agility/Impact**
   a. Medicine ball drops
   b. Weighted hops > 3-way hurdle hop
   c. Hop over hurdle in sagittal plane > hurdle hop into lateral plane
   d. Hop over hurdle (lateral plane) > transition hops
   e. Lateral shuffles, reactive cutting, pro agility drill, T-test drill
   f. Cutting footwork drill > cone cutting > reactive cone cutting
2. **Anterior Chain Strength**
   a. Heavy SL machine extension
   b. Advanced Shuttle Press variations
   c. Leg Extensions on Exxentric
   d. Squat Variations
3. **Functional Strength**
   a. Continue with Phase 5 exercises
   b. Add more load to functional movements
   c. Exxentric: squat, split squat
4. **Posterior Chain Strength**
   a. Hamstring curl progression
   b. Hip thruster progression
   c. Deadlift/RDL variations
   d. Nordic progression
   e. KB swings
   f. Calf raise variations
5. **Lateral Chain Strength**
   a. Lateral step ups
   b. Lateral squat variations
   c. Exxentric: lateral lunges
   d. Rotational deadlifts

## Special Considerations

- Icing & Stretching: At the end of session or at home; Continue prone quad & couch stretch)
- Programming:
  - No > 3-4 sets of 3-5 reps agility
  - 2:1 Power to Strength ratio
  - Undulated Loading Model: Example: Mon - Mod intensity; Wed - High intensity; Fri - High volume
  - Rest Days: At least 3 days off
  - Tempo: 0-0-0 Decelerate the load and transition to acceleration as fast as possible
  - Plyometrics should be done prior to strengthening in order to avoid fatigue
  - Once patient can tolerate advanced plyometrics for multiple sets/reps without pain and with good form you can begin supersetting strength with power
- Conditioning: Begin through cross training on elliptical, bike, rower, jogging

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# Tibial Plateau Fracture Protocol Phase 7

| Purpose | Precautions |
|---|---|
| Return to Sport | |
| **Typical Time Frame** | |
| 6 Months + | |

## Criteria to Pass

- SL hop test >95% LSI with average of 3 trials
- Triple hop test >95% LSI with average of 3 trials
- <10% quad strength deficit for average force with isometric testing at 90° knee flexion
- Girth measurements of the quad within 1 cm of non-surgical limb
- Score of 48 or > on the Injury-Psychological Readiness to Return to Sport (I-PRRS) Questionnaire

## Priority List & Suggested Exercises

1. **Agility/Impact**
   a. Continue with phase 6 agility/impact exercises until performing proficiently
   b. Depth jump progression
2. **Complex Training:** *Compound Exercise at 80% 1RM (4 reps) followed by a plyometric jump exercise that mimic the same motor pattern (3 reps); 3-6 sets*
   a. **Anterior Chain Complex Example:** Front Squats followed by Squat Jumps
   b. **Posterior Chain Complex Example:** Barbell RDLs followedy by KB Swings
   c. **Lateral Chain Complex Example:** Lateral Lunges > Skaters
3. **Contrast Training**
   a. *Perform a maximal or near maximal lift (90-97% 1RM) followed by a drop set performed at 50-70% 1RM. Perform at maximal velocity.*
4. **French Contrast Training**
   a. *4 exercises performed in succession*
   b. Heavy compound exercise (80-90% 1RM) 1-4 reps
   c. Plyometric jump 3 reps
   d. Drop set or weighted jump (30% 1RM) 3 reps
   e. Accelerated jump (usually band assisted) 2-3 reps

## Special Considerations

- Icing: At the end of session or at home
- Prioritize Recovery: 2-3 days off/wk
- Training Scheme: Master complex training > contrast training > french contrast training
- Patients should have about 1 month of sport specific movements before beginning contact. Build confidence with stopping, cutting, training, game like movements/plays, etc.
- Contact should be progressed per therapist discretion during practice. Basketball Example: Contact Drills > 3v3 > 2v2 > 1v1 > 5v5
- Continue building isolated quad strength and explosive movements
- Ramp up conditioning specific to sport

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